NON-WSQ COURSE SLIDES

Data Analytics with R

Course Code: C182

2-day short course · 15 hours

Trainer: Dr Alfred Ang



Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Course Orientation

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About the Trainer



Your Trainer

Complete before class

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

EXPERTISE _____

EXPERIENCE _____

CONTACT _____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and Chief Instructional Designer
EXPERTISE	R, statistics, data analytics, visualisation and AI
CERTIFICATION	ACLP-certified adult educator
DELIVERS	Professional technology programmes
EXPERIENCE	Research, industry analytics and workforce training
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LEARNER INTRODUCTION

Let's Know Each Other

1

Your work

Role, organisation and research or decisions you support.

2

Your R baseline

Importing data, scripts, statistics or ggplot2.

3

Your data challenge

One recurring preparation, analysis or communication problem.

4

Your outcome

What you want to explain more reliably with evidence.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time and attend both days in full.

2. **Participate**
   Ask, test and explain your reasoning.

3. **Respect**
   One conversation; critique ideas, not people.

4. **Protect data**
   Use only the supplied synthetic training datasets.

5. **Save evidence**
   Retain source data, scripts, outputs and run logs.

6. **Learn by doing**
   Run, inspect, challenge and interpret your own result.

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SCHEDULE

Lesson Plan — 2 Days

DAY 1

1. 9:30–10:50 · Data workflow and R setup
2. 10:50–12:40 · Preparation and transformation
3. 1:10–3:00 · Descriptive statistics and summaries
4. 3:00–5:30 · Correlation, regression and testing

DAY 2

1. 9:30–12:20 · Qualitative text analysis
2. 12:50–1:55 · Word clouds and interpretation
3. 1:55–4:55 · Grammar of graphics and visual QA
4. 4:55–5:30 · Course recap and next steps

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COURSE REPOSITORY

Access the Hands-On Labs

Option A · Clone

- Use Git to clone the repository
- Open the numbered lab folder
- Run the .R script from that folder

Option B · Download ZIP

- GitHub > Code > Download ZIP
- Extract the full folder tree
- Keep script, data and workbook together

https://github.com/tertiarycourses/C182-Data-Analytics-with-R

Lab discipline

- Use the LG for detailed procedures
- Preserve source data and control logs
- Complete the Excel evidence workbook

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LMS / TMS

Download Course Material

1 · Sign in

- Open the LMS/TMS portal
- Use your learner account
- Select this course

2 · Download

- Open Courseware
- Download learner slides and guide
- Save beside your lab folders

https://lms-tms.tertiaryinfotech.com/

3 · Verify

- **Check the course code**
- Use the current version
- Ask the trainer if a file differs

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01

TOPIC 01

Data Preparation and Transformation

Research-data workflow · import/export · tidy structure · cleaning · joins · transformation

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Key Concepts — Data Preparation and Transformation

1

Question before code

Define the decision, population, variables and unit of observation before transforming data.

2

Tidy structure

Each variable is a column, each observation is a row and each value occupies one cell.

3

Typed imports

Dates, factors, text and numbers need deliberate types so analysis behaves predictably.

4

Transformation pipeline

Select, filter, mutate, join, reshape and aggregate as visible reproducible stages.

5

Missingness is information

Distinguish structurally absent, unknown and invalid values before removal or imputation.

6

Reconciliation

Prove row counts, keys and material totals before and after every transformation.

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TOPIC 01 · LO1

Research data becomes evidence through a controlled workflow

CONCEPT

1. Frame: Define the research question, population, outcome and unit of observation.
2. Acquire: Record source, owner, period, consent and extraction conditions.

CONTROL / DECISION

1. Prepare: Type, clean, transform and join without destroying source evidence.
2. Validate: Reconcile records and totals before analysis or visualisation.

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R analysis packages form a coherent toolkit

Base R

- Import, inspect, reshape, model and plot with the core language.

tidyverse

- A consistent grammar for import, manipulation and visualisation.

readxl

- Reads Excel worksheets while preserving typed rectangular data.

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Tidy data makes transformations composable

1. Variable — Each measured attribute occupies one column.
2. Observation — Each unit or event occupies one row.
3. Value — Each cell contains one atomic value.
4. Table — Each observational unit forms its own analytical table.

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Import is a data-contract decision

1. **Delimiter**
   Confirm comma, tab or other field boundaries.

2. **Header**
   Use stable unique names that survive refresh.

3. **Type**
   Declare dates, identifiers, numbers and categories deliberately.

4. **Locale**
   Check date order, decimal marks and character encoding.

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CSV and Excel serve different purposes

1. **CSV**
   Portable plain text with one table and no formulas or styles.

2. **Excel**
   Multiple sheets, formulas and formatted control/evidence surfaces.

3. **Risk**
   Spreadsheet display can hide underlying types or errors.

4. **Control**
   Keep an analysis-ready CSV beside the learner evidence workbook.

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Select columns to define analytical scope

CONCEPT

1. Relevance: Keep variables required by the question and validation controls.
2. Privacy: Exclude identifiers or sensitive fields that are not necessary.

CONTROL / DECISION

1. Order: Place keys, dimensions and measures in a reviewable sequence.
2. Lineage: Document fields removed or renamed and why.

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Filter rows with explicit inclusion rules

Condition

- Write the population rule before seeing the result.

Boolean logic

- Use AND, OR and NOT with deliberate parentheses.

Missing values

- NA comparisons require explicit is.na() handling.

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Logical operators define the population

1

AND — Every condition must be true for a row to remain.

2

OR — At least one condition is sufficient.

3

NOT — Excludes a defined condition rather than an informal exception.

4

Audit — Test rules on known boundary records before full use.

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Missingness is a pattern, not merely a nuisance

1. **Unknown**
   The value was not observed or recorded.

2. **Not applicable**
   The variable has no meaning for this observation.

3. **Invalid**
   A value fails type, range or relationship rules.

4. **Decision**
   Exclude, impute or retain with a documented rationale.

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Mutate creates derived analytical variables

1

Formula

Express the relationship in visible code, such as BMI = weight/height².

2

Unit

Check that source units make the result meaningful.

3

Boundary

Test zero, negative, missing and extreme inputs.

4

Trace

Keep the original columns beside the derived value.

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Dates turn periods into ordered evidence

CONCEPT

1. Source: Year and week fields need a documented calendar convention.
2. Parse: Convert strings to true dates instead of plotting labels.

CONTROL / DECISION

1. Validate: Reject impossible or ambiguous dates visibly.
2. Use: True dates enable sorting, intervals, trends and seasonality.

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Joins enrich data—and can corrupt it silently

Left join

- Preserves every analytical record while adding reference attributes.

Inner join

- Keeps only matched records and can hide missing keys.

Many-to-many

- Duplicates records and inflates totals when keys are not unique.

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Join type follows the analytical question

1

Left — Keep all observations from the primary dataset.

2

Inner — Retain only keys present in both tables when justified.

3

Full — Surface every key from both sources for reconciliation.

4

Anti — Isolate unmatched records as a quality exception.

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Reshaping changes structure, not meaning

1. **Long**
   One measurement column plus key/value variables supports grouping and plotting.

2. **Wide**
   Repeated categories occupy columns for comparison or delivery.

3. **Identifier**
   Keys must remain stable through every pivot.

4. **Reconcile**
   Totals and distinct observations should remain explainable.

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Aggregation changes the unit of observation

1. **Group**
   Define the dimensions that create each summary row.

2. **Measure**
   Choose sum, count, mean or another statistic intentionally.

3. **Denominator**
   Retain n and exposure when comparing groups.

4. **Risk**
   Aggregates can conceal within-group variation and missingness.

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Pipes make the analytical story readable

CONCEPT

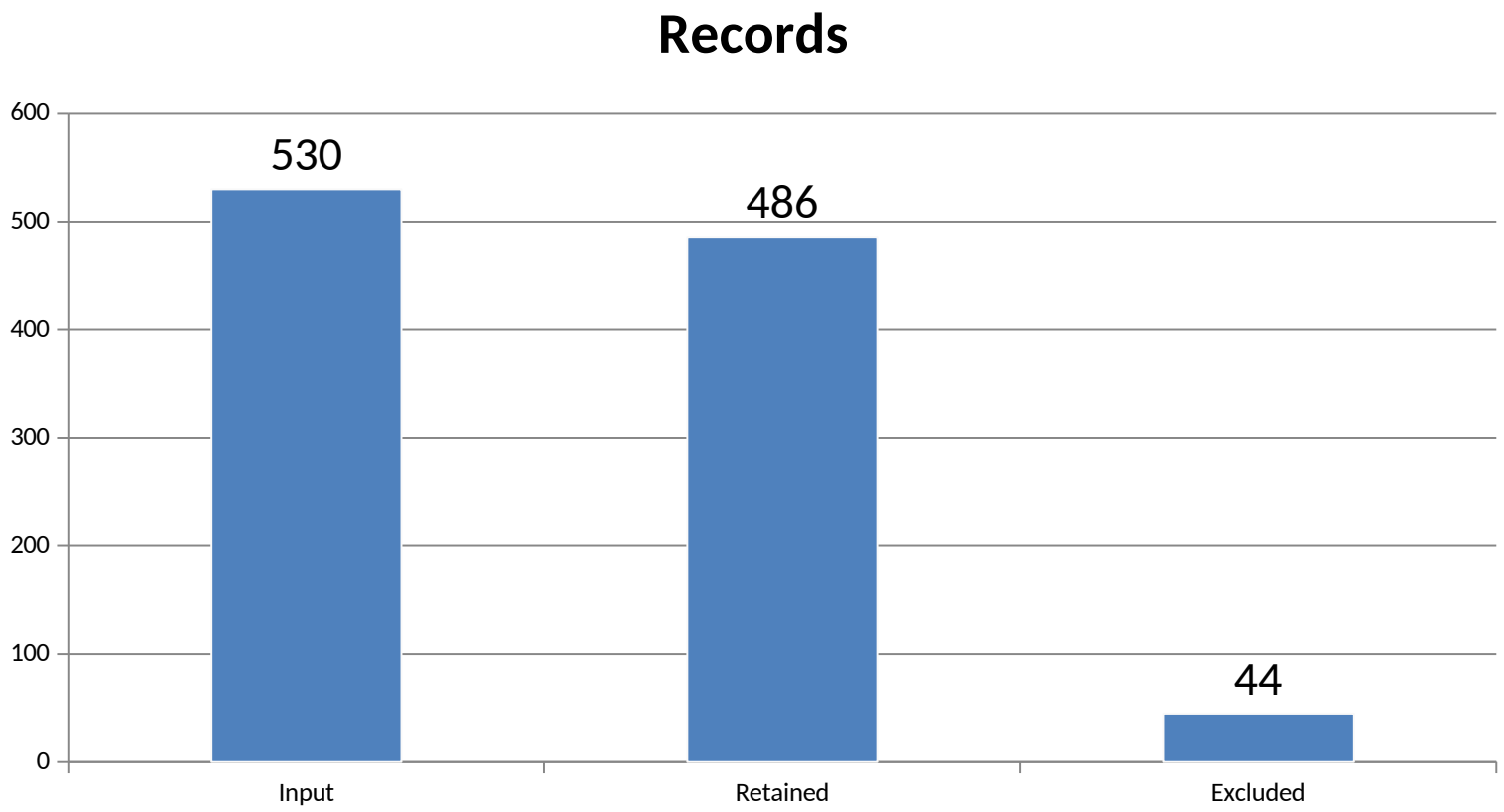
1. Sequence: Each operation receives the result of the previous stage.
2. Names: Assign meaningful intermediate outputs at review boundaries.

CONTROL / DECISION

1. Debug: Inspect after import, join, filter and aggregation gates.
2. Reproducibility: The same source and code should recreate the result.

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Reconciliation catches silent loss



EVIDENCE LENS

A clean output is credible only when retained and excluded records reconcile to the input.

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A release-ready dataset carries evidence

1. Source — Origin, extraction date and owner.
2. Code — Versioned script and package/session information.
3. Controls — Row, key, total and exception checks.
4. Limitations — Known missingness, bias, transformations and allowed use.

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Prepare and Transform Dengue Surveillance Data

LAB 1

Import weekly dengue surveillance data, inspect types and missingness, filter a defined population, construct dates and export a controlled analysis table.

EXPECTED OUTPUT

A reproducible clean dengue time-series dataset, transformation log and reconciliation evidence.

TOOLS R, read.csv, dplyr-compatible workflow, date construction, write.csv

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LAB 1 · ACCEPTANCE

Prepare and Transform Dengue Surveillance Data

ACCEPTANCE CHECK

The script runs without error, output rows reconcile to input less documented exclusions, dates are valid, and the clean CSV and plot are created.

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Join, Reshape and Aggregate Vaccination Data

LAB 2

Standardise vaccination and clinic reference data, join on controlled keys, reshape between wide and long forms and produce an auditable annual summary.

EXPECTED OUTPUT

A joined long-form vaccination dataset, unmatched-key report, annual summary and transformation audit.

TOOLS R, merge, reshape, aggregate, key validation, reconciliation

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LAB 2 · ACCEPTANCE

Join, Reshape and Aggregate Vaccination Data

ACCEPTANCE CHECK

The left join preserves all source rows, unmatched keys are zero or documented, and total doses reconcile across input, long and wide summaries.

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TOPIC RECAP

Recap — Data Preparation and Transformation

1. **Knowledge**
   Question before code and related principles.

2. **Ability**
   LO1

3. **Evidence**
   A validated workbook and decision-ready interpretation.

4. **Next**
   Connect this topic to the next layer of the analytical workflow.

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02

TOPIC 02

Data Summary

Data types · descriptive statistics · distribution · grouped summaries · basic plots and tables

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Key Concepts — Data Summary

1

Variable type drives method

Nominal, ordinal, discrete and continuous data support different summaries and charts.

2

Centre

Mean and median answer different questions when the distribution is skewed or contains outliers.

3

Spread

Range, interquartile range, variance and standard deviation quantify variability.

4

Grouped summaries

Compare meaningful sub-populations while preserving denominators and sample sizes.

5

Sampling variability

A statistic changes from sample to sample; uncertainty is part of the result.

6

Review accuracy

Recompute, reconcile and inspect distributions before accepting a summary.

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Variable type determines valid summaries

CONCEPT

1. Nominal: Categories with no inherent order.
2. Ordinal: Categories with an order but not equal intervals.

CONTROL / DECISION

1. Discrete: Countable numeric outcomes.
2. Continuous: Measurements on an interval or ratio scale.

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Qualitative and quantitative answer different questions

Qualitative

- What kinds, themes or experiences occur?

Quantitative

- How many, how much or how strongly?

Mixed evidence

- Counts locate patterns; text explains context and meaning.

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Frequency tables reveal category structure

1

Count — Number of observations in each level.

2

Proportion — Count divided by a clearly stated denominator.

3

Missing — Display unknown or excluded categories explicitly.

4

Comparison — Use common denominators when comparing groups.

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Mean describes balance; median describes position

1

Mean

Uses every value and is sensitive to extremes.

2

Median

Splits ordered observations into two halves.

3

Symmetry

Mean and median are usually close.

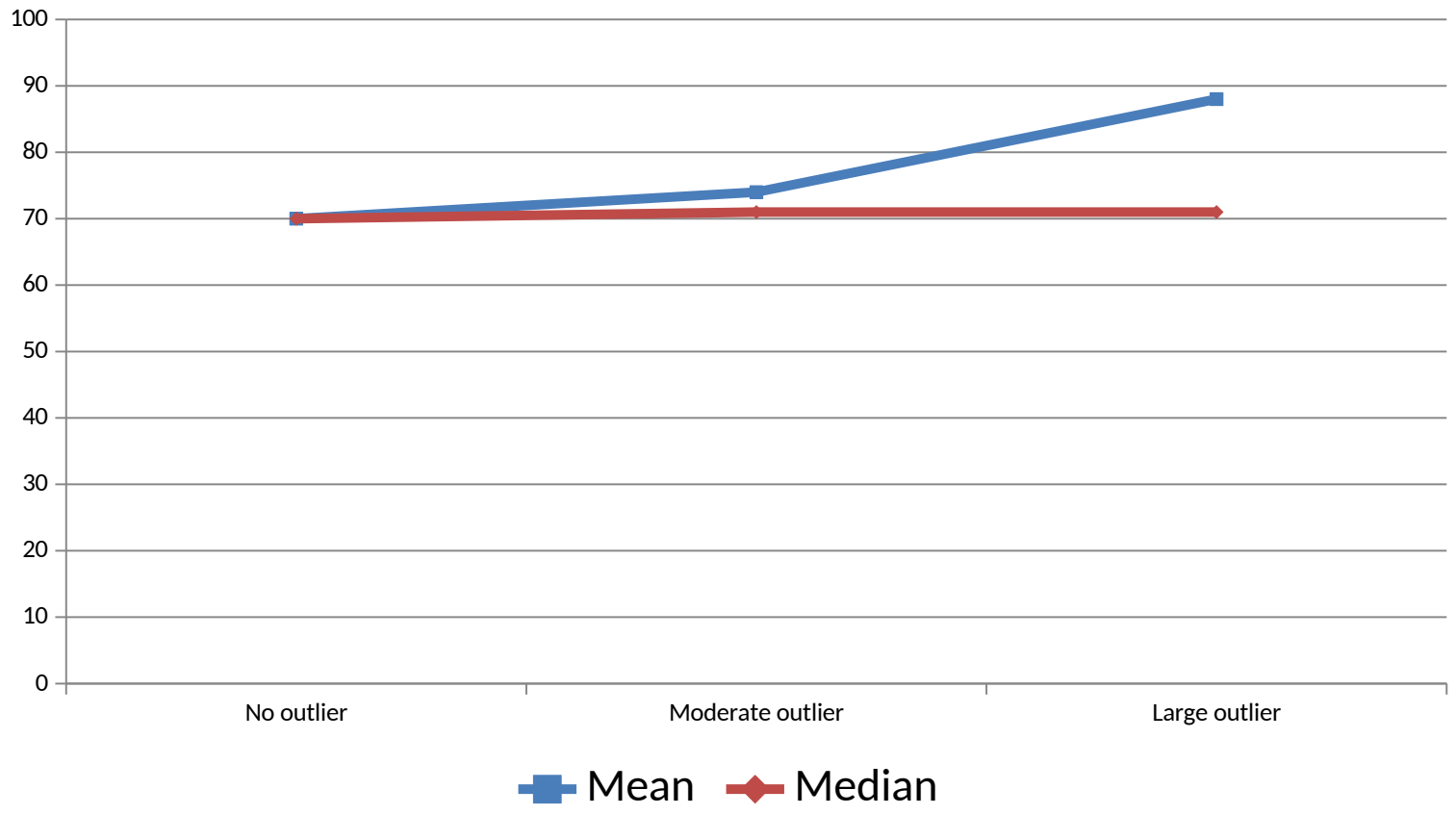
4

Skew

Their separation is a warning to inspect distribution.

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Mean and median respond differently to outliers



EVIDENCE LENS

The mean moves with extreme values; the median remains anchored to the ordered middle.

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Trimmed means trade sensitivity for robustness

CONCEPT

1. Rule: Remove a fixed proportion from each tail before averaging.
2. Benefit: Reduces the influence of extreme observations.

CONTROL / DECISION

1. Cost: Discarded tails may contain real, important cases.
2. Use: Compare trim levels rather than selecting one after the fact.

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Range is simple but fragile

Definition

- Maximum minus minimum.

Strength

- Easy to calculate and explain.

Weakness

- Depends on only two observations.

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Variance and standard deviation quantify spread

1

Variance — Average squared deviation, estimated with n−1 for a sample.

2

SD — Square root of variance in the original unit.

3

Meaning — Larger SD means observations are more dispersed around the mean.

4

Limit — Strong skew or outliers can make SD hard to interpret alone.

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The IQR focuses on the middle half

1. **Q1**
   25% of observations lie at or below this value.

2. **Q3**
   75% lie at or below this value.

3. **IQR**
   Q3 − Q1, resistant to tail extremes.

4. **Outlier rule**
   1.5 × IQR flags values for investigation, not automatic deletion.

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A five-number summary sketches the distribution

1. **Minimum**
   Lowest observed value.

2. **Quartiles**
   Q1, median and Q3 locate the middle structure.

3. **Maximum**
   Highest observed value.

4. **Review**
   Compare with units, n and possible data-entry errors.

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Histograms depend on bin choices

CONCEPT

1. Bins: Intervals convert continuous values into counts.
2. Shape: Reveal skew, modes, gaps and tails.

CONTROL / DECISION

1. Sensitivity: Different widths can hide or exaggerate patterns.
2. Control: Use a defensible rule and show the underlying range.

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Box plots make group comparison compact

Box

- Middle 50% from Q1 to Q3.

Line

- Median within the box.

Whiskers

- Typical range under the chosen convention.

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Grouped summaries require denominators

1. n — Shows how much evidence supports each group statistic.
2. Missing — State whether excluded cases differ by group.
3. Unequal groups — Precision and visual weight differ with sample size.
4. Aggregation — Do not compare sums where rates or means answer the question.

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Sampling links observed data to a population

1. **Population**
   All units the research decision concerns.

2. **Sample**
   Observed subset used to estimate population characteristics.

3. **Parameter**
   Unknown population quantity.

4. **Statistic**
   Sample-based estimate that varies across samples.

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Sampling distributions explain uncertainty

1. **Repeated samples**
   The same statistic takes different values.

2. **Standard error**
   Expected variability of an estimate across samples.

3. **Sample size**
   Larger n usually improves precision, not representativeness.

4. **Bias**
   Systematic selection errors remain even with more data.

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Accuracy review precedes interpretation

CONCEPT

1. Recompute: Check a material statistic by an independent method.
2. Reconcile: Counts and grouped totals agree with source data.

CONTROL / DECISION

1. Inspect: Plots reveal impossible values or unexpected shapes.
2. Document: Record exclusions, rounding, units and package versions.

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Summary tables should carry meaning

Title

- State population, measure and period.

Units

- Keep kg, cases, percent and counts explicit.

Precision

- Round only for communication; retain underlying values.

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A defensible summary separates signal from caveat

1. Pattern — What centre, spread or group difference is observed?
2. Magnitude — How large is it in the original unit?
3. Uncertainty — How much evidence and variability surround it?
4. Limit — What should the reader not conclude from the summary?

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Summarise Centre, Spread and Distribution

LAB 3

Compare mean, trimmed mean and median, quantify variability and inspect how outliers affect health-service waiting-time summaries.

EXPECTED OUTPUT

A descriptive-statistics table, distribution plot, outlier review and accuracy checklist.

TOOLS R, mean, median, sd, IQR, quantile, hist, boxplot

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Summarise Centre, Spread and Distribution

ACCEPTANCE CHECK

R and Excel checks agree, units and sample size are stated, outliers remain traceable, and the chosen centre measure matches the observed distribution.

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Build Grouped Tables and Basic Plots

LAB 4

Summarise health observations by category, preserve denominators and review the accuracy of proportions, grouped means and basic plots.

EXPECTED OUTPUT

Frequency/proportion tables, grouped numerical summaries, bar and box plots, and a denominator audit.

TOOLS R, table, prop.table, aggregate, barplot, boxplot

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LAB 4 · ACCEPTANCE

Build Grouped Tables and Basic Plots

ACCEPTANCE CHECK

Counts reconcile to the dataset, proportions sum to 100% within rounding, every grouped statistic includes n, and the written comparison matches both tables and plots.

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TOPIC RECAP

Recap — Data Summary

1

Knowledge

Variable type drives method and related principles.

2

Ability

LO2

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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03

TOPIC 03

Quantitative Data Analysis

Correlation · regression · hypothesis testing · effect size · ANOVA · interpretation

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Key Concepts — Quantitative Data Analysis

1

Correlation

Measures strength and direction of association; it does not by itself establish causation.

2

Regression

Models an outcome as a function of one or more predictors under explicit assumptions.

3

Hypothesis testing

Compares observed evidence with a pre-specified null model and significance threshold.

4

Effect size

Quantifies how large the difference or relationship is in practical terms.

5

ANOVA

Tests whether group means differ beyond what within-group variability would suggest.

6

Interpretation boundary

Separate evidence, assumptions, uncertainty and business or research judgement.

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Quantitative analysis begins with a model of the question

CONCEPT

1. Outcome: The response to explain, compare or predict.
2. Predictor: A measured factor believed to relate to the outcome.

CONTROL / DECISION

1. Design: Paired, independent, observational or experimental.
2. Assumptions: Conditions required for valid estimation or testing.

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Covariance and correlation describe co-movement

Covariance

- Direction and joint variability in combined units.

Correlation

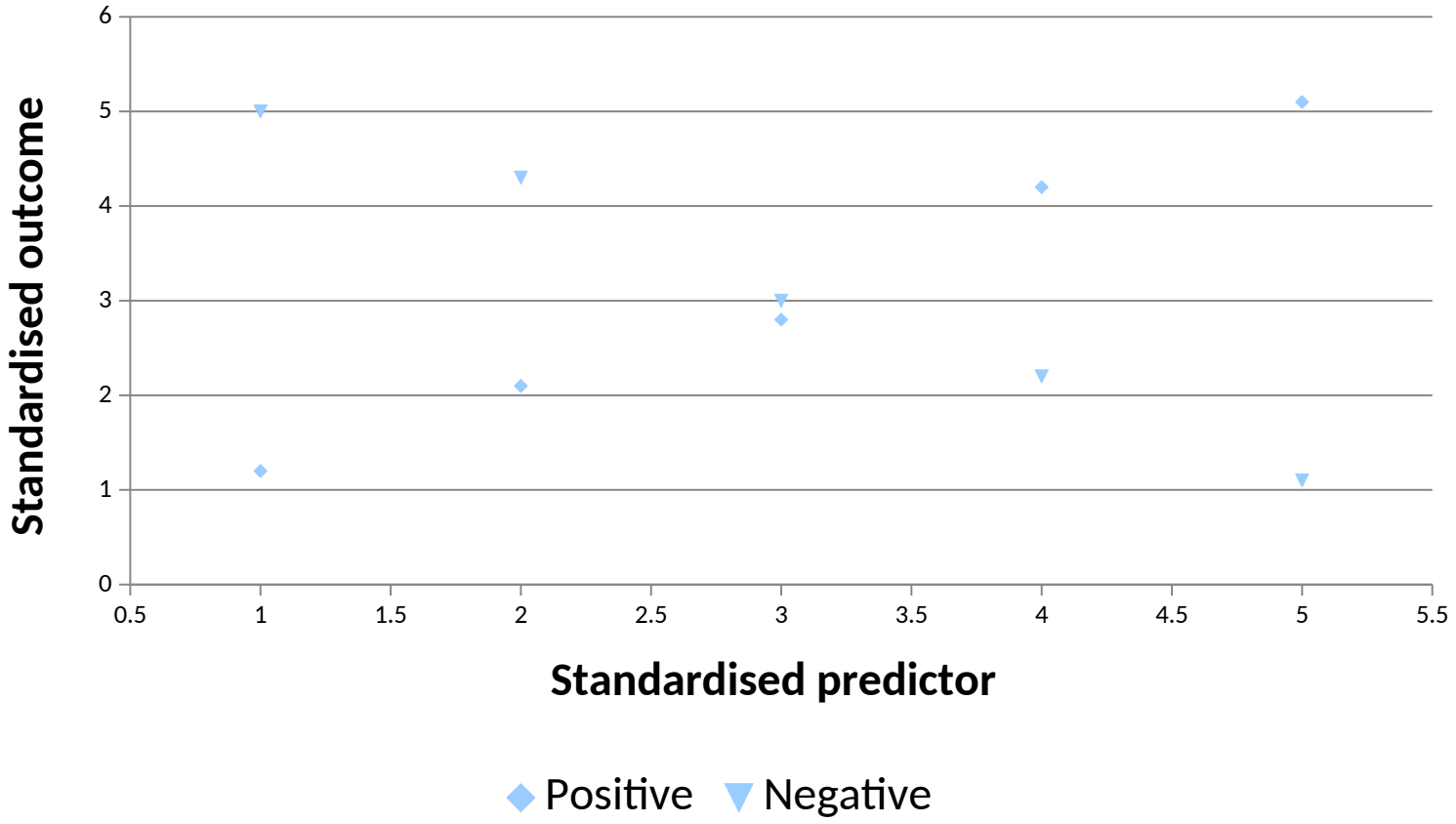
- Standardised direction and strength from −1 to +1.

Positive

- Variables tend to increase together.

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Correlation strength and direction



EVIDENCE LENS

Direction and linear strength are visible, but neither pattern proves causation.

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Pearson and rank correlation make different assumptions

1. **Pearson**
   Measures linear association between quantitative variables.

2. **Spearman**
   Measures monotonic association using ranks.

3. **Choice**
   Inspect form, scale and outliers before selecting.

4. **Report**
   Coefficient, interval, p-value, n and a scatter plot.

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Correlation is not causation

1. **Confounding**
   A third factor may drive both variables.

2. **Reverse direction**
   The assumed predictor may actually respond to the outcome.

3. **Selection**
   Observed units may not represent the population.

4. **Evidence**
   Causal claims require design, timing and alternative explanations.

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Linear regression expresses an average relationship

CONCEPT

1. Intercept: Expected outcome when predictors equal zero.
2. Slope: Expected outcome change per unit of predictor.

CONTROL / DECISION

1. Error: Observed minus modelled outcome.
2. Fit: Estimated line minimises squared residuals.

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Regression coefficients need units and conditions

Direction

- Positive or negative adjusted relationship.

Magnitude

- Change in outcome units per predictor unit.

Condition

- Other included predictors held constant.

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R-squared measures explained variation

1

Range — Usually 0 to 1 for models with an intercept.

2

Meaning — Share of outcome variation explained in this sample.

3

Adjusted — Penalises predictors that add little information.

4

Limit — High fit does not prove causation or future accuracy.

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Residuals reveal model mismatch

1. **Randomness**
   No systematic curve or pattern across fitted values.

2. **Spread**
   Variability should not fan out dramatically.

3. **Outliers**
   Large residuals require context and sensitivity review.

4. **Dependence**
   Time or clusters can violate independence.

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Hypothesis testing is a pre-specified decision rule

1. **H0**
   Baseline model or no-effect statement.

2. **H1**
   Difference or direction the study is designed to detect.

3. **Alpha**
   Chosen false-positive risk threshold.

4. **Evidence**
   Test statistic and p-value under H0 assumptions.

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P-values have a narrow meaning

CONCEPT

1. Conditional: Calculated assuming H0 and the test assumptions.
2. Extremeness: Probability of data at least this extreme under H0.

CONTROL / DECISION

1. Not effect: Does not tell how large or useful the finding is.
2. Not truth: Is not the probability that H0 is true.

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One- and two-tailed tests answer different questions

Two-tailed

- Detects differences in either direction.

One-tailed

- Only a justified pre-specified direction counts.

Integrity

- Do not choose the tail after observing the result.

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Type I and Type II errors trade off

1

Type I — Reject a true H0: false positive.

2

Type II — Fail to reject a false H0: false negative.

3

Power — Probability of detecting a meaningful true effect.

4

Design — Sample size, variability and effect magnitude affect both.

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T-tests must match how data was collected

1. **One sample**
   Compare a sample mean with a benchmark.

2. **Independent**
   Compare two separate groups.

3. **Paired**
   Compare matched units or repeated measures.

4. **Welch**
   Allows unequal group variances and is often safer.

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Effect size complements significance

1. **Mean difference**
   Magnitude in the original unit.

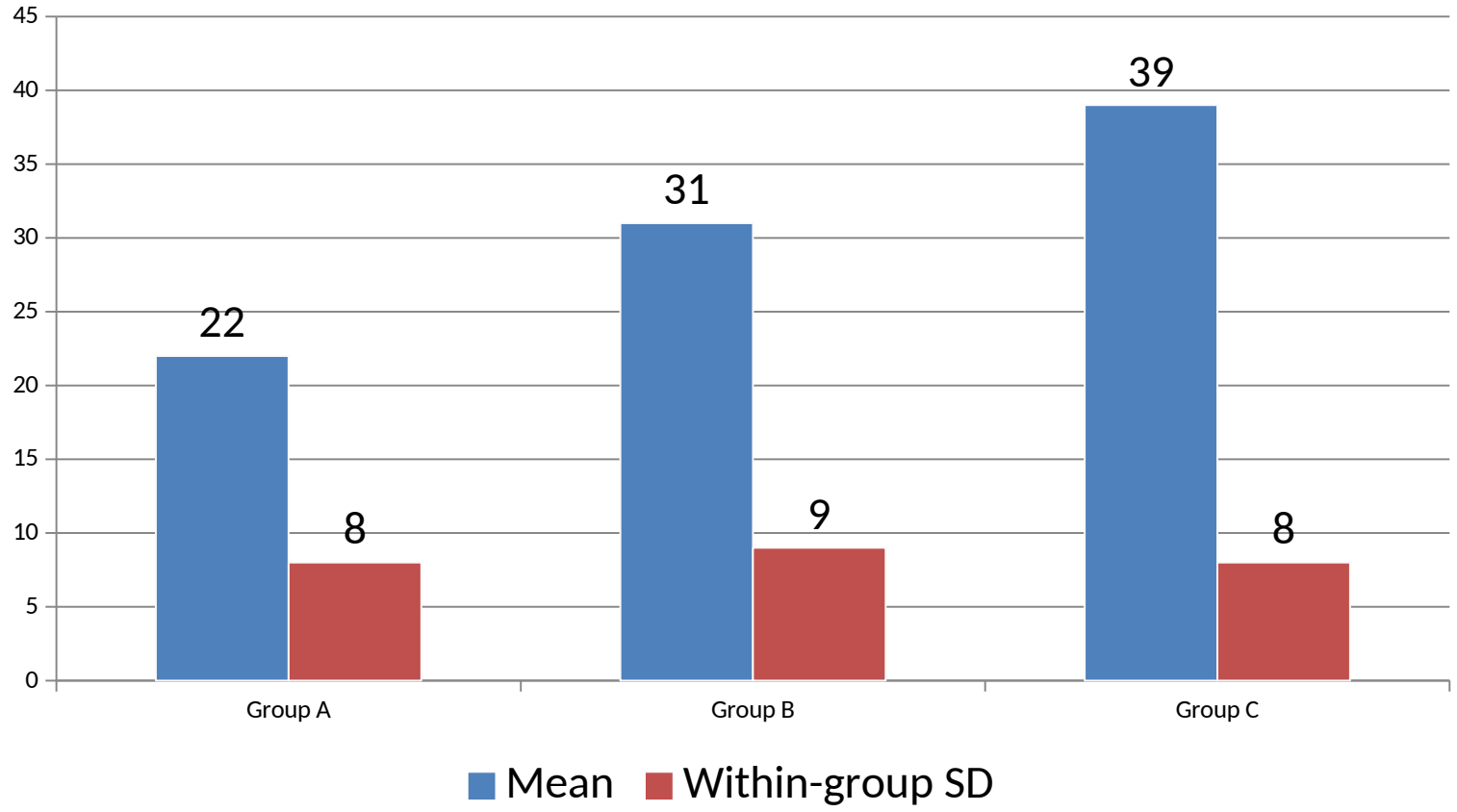
2. **Standardised**
   Difference scaled by variability.

3. **Interval**
   Range of plausible effect values under assumptions.

4. **Decision**
   Compare with cost, risk and practical threshold.

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ANOVA separates within and between variation



EVIDENCE LENS

ANOVA asks whether between-group separation is large relative to within-group variability.

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Assumptions are evidence, not a checkbox

Independence

- Comes primarily from study design.

Distribution

- Inspect residuals or within-group shapes.

Variance

- Review group spreads; use robust alternatives when needed.

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Interpretation is a structured argument

1. Estimate — State direction and magnitude.
2. Uncertainty — Give interval, variability and sample size.
3. Evidence — Report test/model and p-value without overclaim.
4. Limit — Name assumptions, design constraints and next evidence.

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Analyse Correlation and Regression

LAB 5

Examine a quantitative relationship, fit a linear model, evaluate residuals and communicate an interpretation without causal overclaiming.

EXPECTED OUTPUT

Correlation evidence, fitted regression model, diagnostic plot, prediction and limitations statement.

TOOLS R, cor, cor.test, lm, summary, predict, residual diagnostics

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Analyse Correlation and Regression

ACCEPTANCE CHECK

The model output is reproducible, coefficient and R-squared interpretations use correct units, diagnostics are reviewed, and the conclusion avoids causal claims.

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Test Differences and Compare Groups

LAB 6

Pre-specify hypotheses, run a two-sample t-test and one-way ANOVA, quantify effects and separate statistical from practical significance.

EXPECTED OUTPUT

Test-design record, t-test and ANOVA outputs, effect estimates, assumption checks and conclusion.

TOOLS R, t.test, aov, TukeyHSD, effect-size calculation, boxplot

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Test Differences and Compare Groups

ACCEPTANCE CHECK

Test choices match the designs, hypotheses were pre-specified, effect estimates accompany p-values, and conclusions state assumptions and practical limitations.

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TOPIC RECAP

Recap — Quantitative Data Analysis

1

Knowledge

Correlation and related principles.

2

Ability

LO3

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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Lunch Break

30 minutes

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04

TOPIC 04

Qualitative Data Analysis

Text preparation · tokenisation · stop words · term frequency · word clouds · thematic interpretation

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Key Concepts — Qualitative Data Analysis

1

Qualitative evidence

Open-ended responses carry meaning, context and ambiguity that numeric summaries can miss.

2

Text preparation

Normalise case and punctuation while preserving the source and recording every exclusion.

3

Tokens

Break text into reproducible analytical units such as words, phrases or coded themes.

4

Stop words

Remove high-frequency low-information terms using a documented list and domain judgement.

5

Frequency is not importance

Common terms reveal salience but require context, concordance and representative excerpts.

6

Credible interpretation

Use a codebook, peer review and traceable excerpts to manage subjectivity.

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Qualitative data preserves experience and context

CONCEPT

1. Open responses: Capture language respondents choose themselves.
2. Themes: Recurring meanings can be coded and compared.

CONTROL / DECISION

1. Context: The same word can carry different meanings in different excerpts.
2. Responsibility: Protect identity and avoid decontextualised quotation.

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A qualitative workflow remains auditable

Preserve

- Keep original response and stable record ID.

Prepare

- Create derived cleaned text without overwriting source.

Code

- Apply documented tokens, themes or categories.

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Text normalisation changes the evidence

1

Case — Lowercase merges capitalisation variants.

2

Punctuation — Removal can change contractions, hashtags or medical codes.

3

Spelling — Correction can merge meaning or introduce analyst judgement.

4

Audit — Record each rule and retain the original text.

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Tokenisation defines the analytical unit

1. **Word**
   Simple and interpretable, but loses phrase meaning.

2. **N-gram**
   Preserves short phrases and negation better.

3. **Sentence**
   Supports context and sentiment review.

4. **Theme**
   Human-coded concept may be more meaningful than a token.

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Stop words reduce noise—and can remove meaning

1. **General**
   Common function words often add little topical information.

2. **Domain**
   Frequently used terms may still be important in healthcare.

3. **Negation**
   Removing not or no can reverse meaning.

4. **Sensitivity**
   Compare results before and after the list changes.

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Term frequency is a starting point

CONCEPT

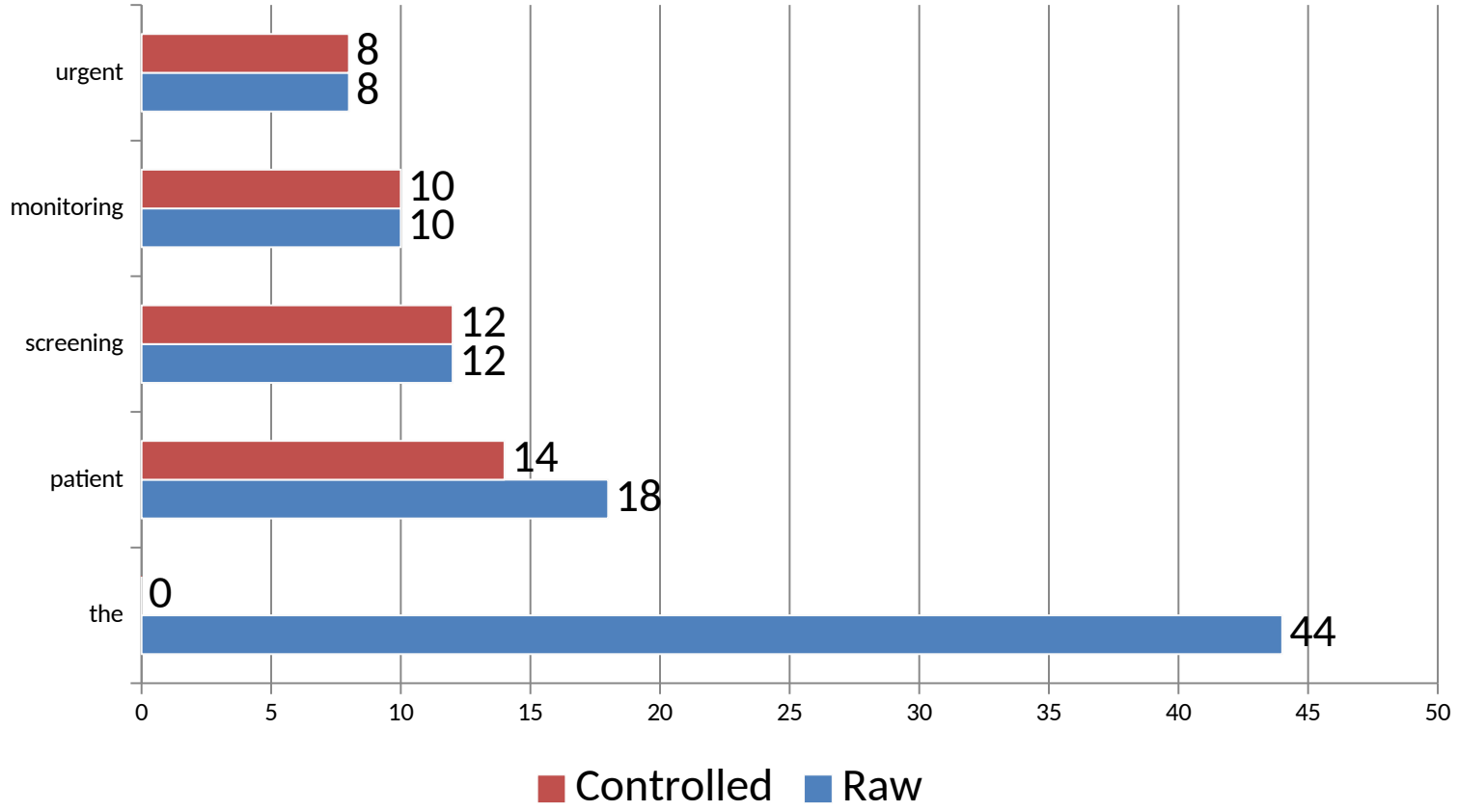
1. Count: How often a token appears.
2. Document frequency: How many responses contain the token.

CONTROL / DECISION

1. Rate: Count relative to total words or responses.
2. Limit: Frequency alone does not establish sentiment, severity or importance.

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Term frequency changes after stop-word control



EVIDENCE LENS

Preprocessing choices materially change visual prominence and must be disclosed.

Editable native PowerPoint chart · illustrative data

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Concordance restores local context

1. Keyword — Choose a term suggested by frequency or theory.
2. Window — Inspect neighbouring words or full response.
3. Meaning — Separate different uses of the same term.
4. Trace — Retain record ID without exposing personal identity.

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A codebook turns interpretation into a process

1. **Code**
   Short label for a defined meaning.

2. **Definition**
   What is included and excluded.

3. **Example**
   Representative excerpt illustrating the code.

4. **Decision rule**
   How ambiguous cases are resolved.

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Human coding needs consistency checks

1. **Calibration**
   Coders discuss examples before full coding.

2. **Double-code**
   Independent review reveals disagreements.

3. **Resolution**
   Record how differences were reconciled.

4. **Reflexivity**
   Acknowledge how analyst perspective shapes interpretation.

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Word clouds are overview visuals, not analysis

CONCEPT

1. Size: Usually encodes term frequency.
2. Strength: Rapidly surfaces prominent vocabulary.

CONTROL / DECISION

1. Weakness: Poor for precise comparison and hides context.
2. Companion: Always provide a ranked table and excerpts.

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A reproducible word cloud fixes random choices

Seed

- Keeps the layout stable across reruns.

Threshold

- State minimum frequency and maximum words.

Palette

- Use readable, accessible colours.

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Preprocessing sensitivity is an effect

1. Choice — Stop words, stemming and token rules change counts.
2. Compare — Measure rank and frequency shifts.
3. Interpret — Claims robust to sensible choices are more credible.
4. Report — Disclose choices with material impact.

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Stemming and lemmatisation trade detail for grouping

1. **Stem**
   Cuts words to a common root, sometimes awkwardly.

2. **Lemma**
   Maps variants to a dictionary form.

3. **Benefit**
   Combines related grammatical forms.

4. **Risk**
   Can merge meanings that should remain distinct.

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Qualitative significance is not only a p-value

1. **Salience**
   Prominence within participant accounts.

2. **Coverage**
   Breadth across responses or channels.

3. **Contrast**
   Meaningful difference between contexts.

4. **Impact**
   Potential consequence supported by evidence and judgement.

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Triangulation strengthens interpretation

CONCEPT

1. Counts: Show prevalence or distribution.
2. Excerpts: Show meaning and context.

CONTROL / DECISION

1. Groups: Compare channels, time or participant categories.
2. Peer review: Challenge selective reading and alternative explanations.

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Qualitative visuals need ethical boundaries

Privacy

- Remove identifiable or sensitive details.

Representation

- Do not let vivid quotes stand in for all respondents.

Proportionality

- Avoid implying population prevalence from convenience feedback.

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A credible theme statement carries evidence and restraint

1. Theme — Name the recurring idea precisely.
2. Support — Provide counts, coverage and representative excerpts.
3. Variation — Show exceptions or contrasting accounts.
4. Limit — State sampling, coding and contextual constraints.

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Prepare Text and Analyse Term Frequency

LAB 7

Convert feedback text into a traceable token table, remove documented stop words and compare term frequencies across service channels.

EXPECTED OUTPUT

Clean token table, stop-word audit, frequency table, channel comparison and interpretation notes.

TOOLS R, regular expressions, strsplit, stop-word filtering, table, aggregate

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Prepare Text and Analyse Term Frequency

ACCEPTANCE CHECK

Every token remains traceable to a source record, exclusions are documented, frequency totals reconcile, and the interpretation uses context rather than frequency alone.

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Create and Critique a Word Cloud

LAB 8

Create a reproducible word cloud or frequency display, test the effect of preprocessing choices and triangulate the visual with representative excerpts.

EXPECTED OUTPUT

Word-cloud image, ranked-frequency table, sensitivity comparison, excerpt evidence and limitations statement.

TOOLS R, wordcloud optional package, base-graphics fallback, RColorBrewer optional

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Create and Critique a Word Cloud

ACCEPTANCE CHECK

The visual is reproducible, ranked counts are available, preprocessing sensitivity is documented, and every claim is supported by counts plus traceable excerpts.

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TOPIC RECAP

Recap — Qualitative Data Analysis

1

Knowledge

Qualitative evidence and related principles.

2

Ability

LO4

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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05

TOPIC 05

Data Visualization

Grammar of graphics · quantitative and qualitative plots · customisation · accessibility · findings

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Key Concepts — Data Visualization

1

Question-to-chart fit

Choose position, length, colour and shape to support the analytical purpose.

2

Grammar of graphics

Build a plot from data, aesthetics, geoms, statistical transforms, scales, coordinates and facets.

3

Truthful scales

Units, baselines, aspect ratios and uncertainty must not exaggerate or hide evidence.

4

Accessible design

Use readable labels, sufficient contrast and redundant encodings beyond colour alone.

5

Layered emphasis

Guide attention with hierarchy and annotation while keeping the complete context available.

6

Interpret findings

State the pattern, evidence, uncertainty, decision relevance and limitation.

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A visualisation answers a question

CONCEPT

1. Trend: How does a measure change over ordered time?
2. Comparison: Which categories are higher, lower or different?

CONTROL / DECISION

1. Distribution: How are values spread and where are unusual observations?
2. Relationship: How do two quantitative variables move together?

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The grammar of graphics separates decisions

Data

- The analytical table and variables.

Aesthetics

- Mappings to x, y, colour, size, shape or group.

Geom

- Marks such as point, line, bar or box.

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Data and aesthetics define the visual contract

1

x/y — Position is usually the most precise quantitative encoding.

2

Colour — Useful for grouping but weak without redundant cues.

3

Size — Area judgements are less precise than position or length.

4

Group — Controls which observations connect or aggregate together.

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Geoms must match the observation and question

1. **Point**
   One mark per paired quantitative observation.

2. **Line**
   Connects ordered observations in a meaningful sequence.

3. **Bar**
   Compares category values using aligned lengths.

4. **Box**
   Summarises a quantitative distribution by group.

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Statistical transforms create derived displays

1. **Count**
   Bars can count records by category.

2. **Bin**
   Histograms group continuous values into intervals.

3. **Smooth**
   A fitted trend summarises relationship form.

4. **Summary**
   Intervals and point estimates reduce data visibly.

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Scales translate data into perception

CONCEPT

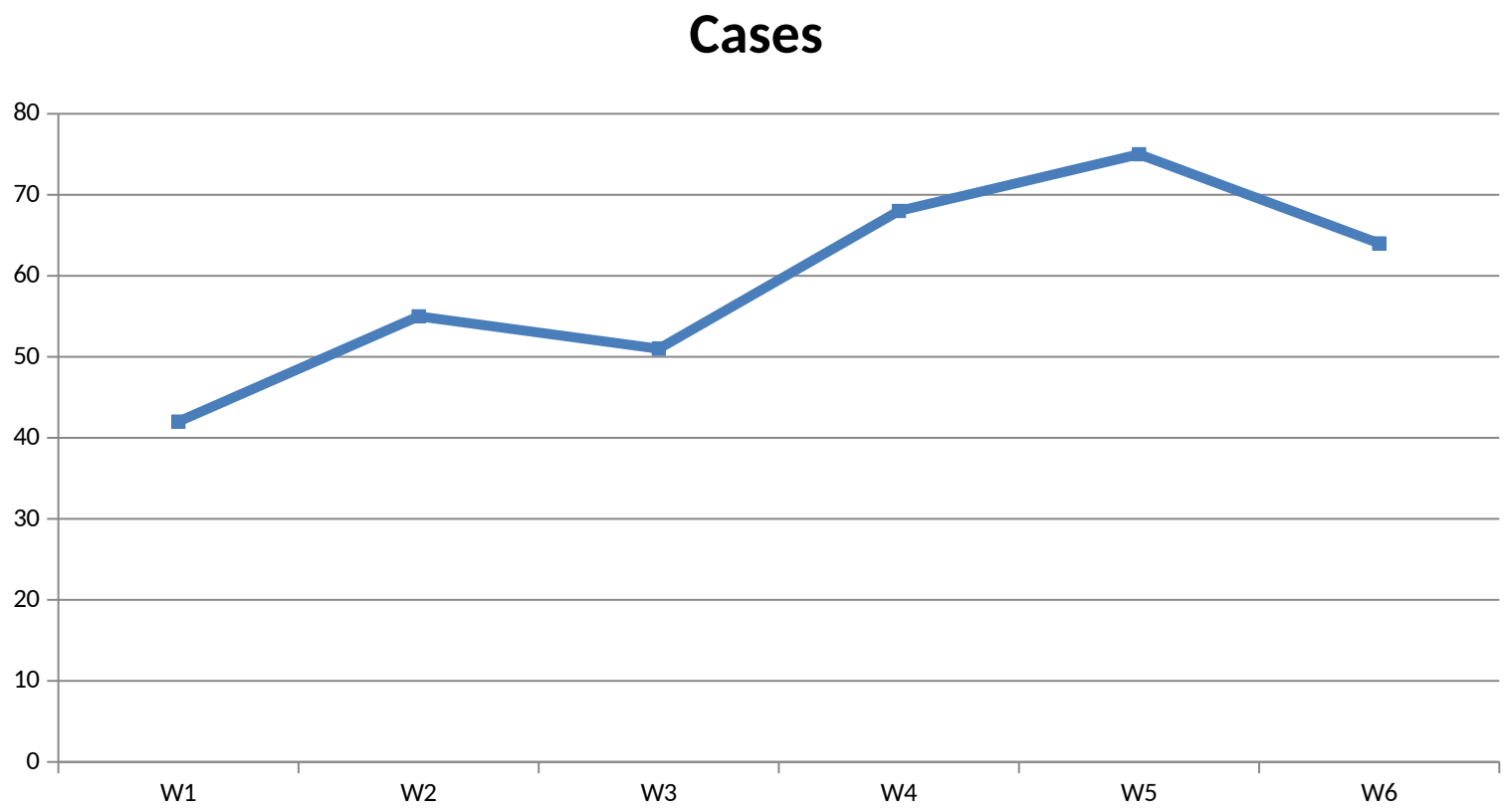
1. Domain: Values included in the mapping.
2. Range: Positions, colours or sizes used on screen.

CONTROL / DECISION

1. Breaks: Tick marks guide interpretation.
2. Labels: Units and formats make values legible.

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Trend visual: signal over time



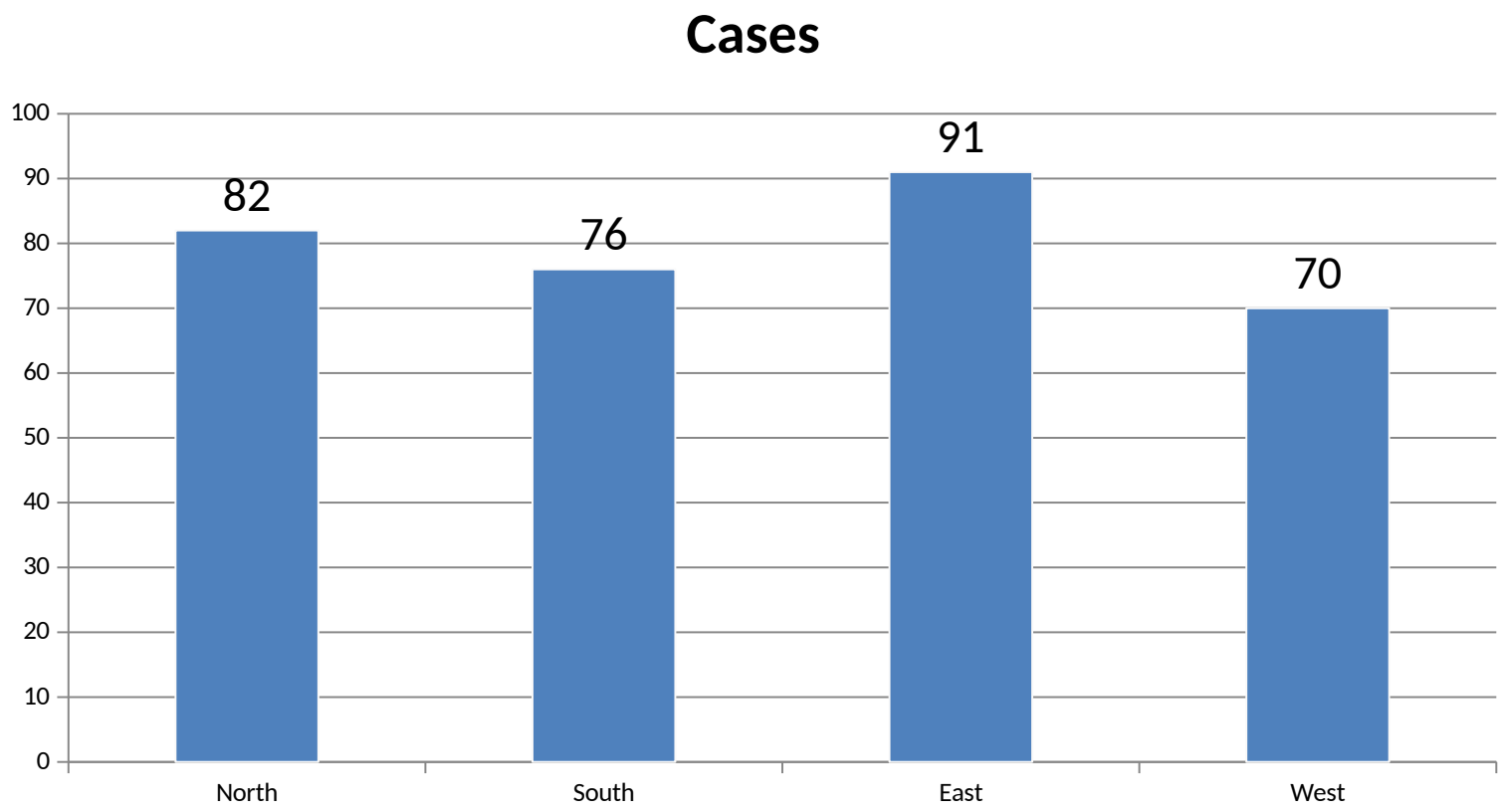
EVIDENCE LENS

Ordered time supports connected marks; annotate meaningful changes without hiding variability.

Editable native PowerPoint chart · illustrative data

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Comparison visual: zero baseline matters



EVIDENCE LENS

Bars encode magnitude by length, so the baseline should start at zero.

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Distribution visuals reveal what averages hide

1. **Histogram**
   Shape across numeric intervals.

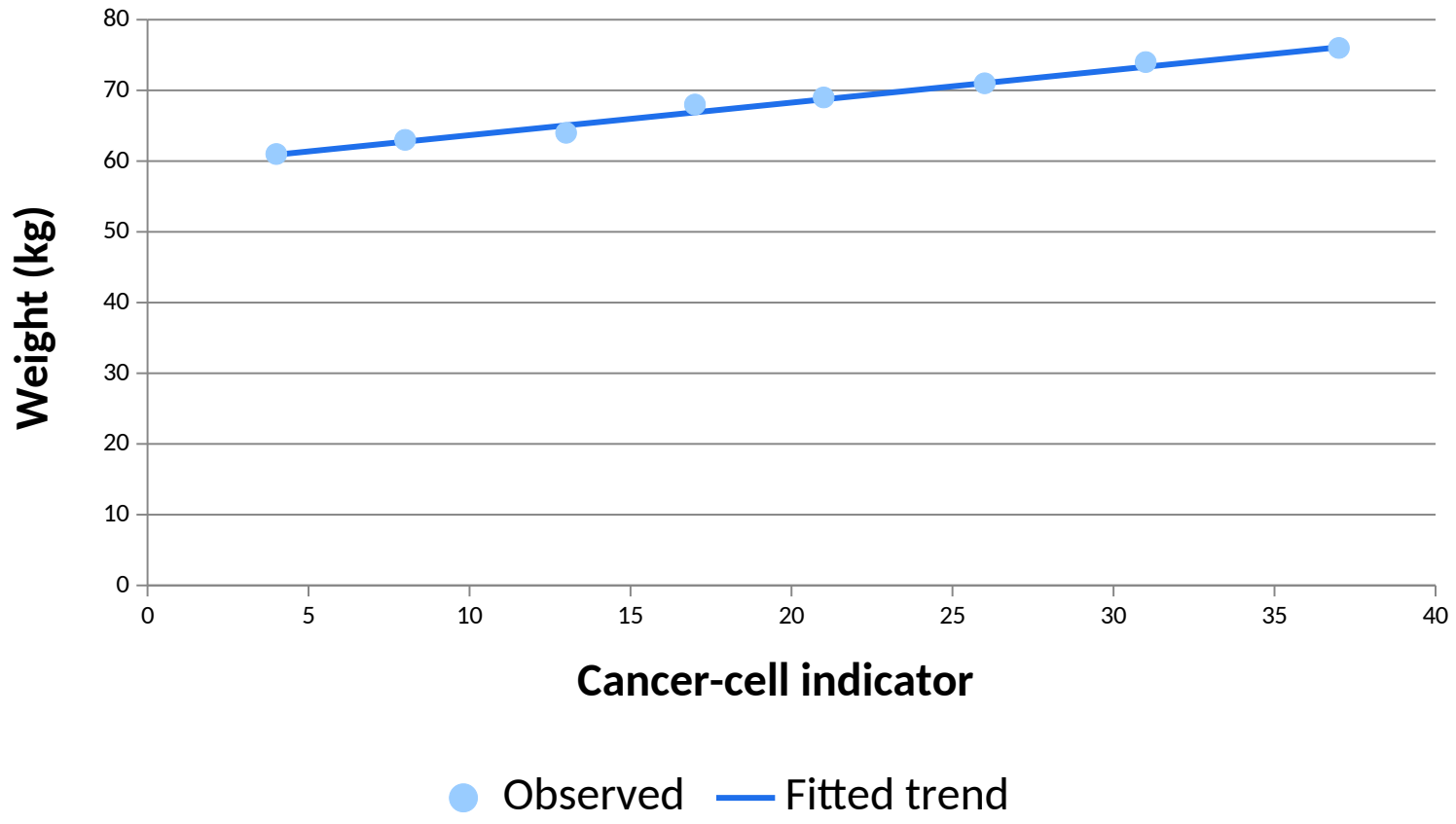
2. **Box plot**
   Compact group comparison of median, IQR and outliers.

3. **Density**
   Smoothed shape sensitive to bandwidth.

4. **Raw points**
   Show sample size and clusters where feasible.

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Quantitative relationship: scatter plus trend



EVIDENCE LENS

A fitted trend can summarise association while individual points retain spread and uncertainty.

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Qualitative plots need precise companions

CONCEPT

1. Ranked bars: Best for comparing term or theme frequencies.
2. Word cloud: Useful only as an overview of prominence.

CONTROL / DECISION

1. Heat map: Shows frequency across groups or time.
2. Excerpts: Restore meaning behind counts and colours.

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Facets create small multiples

Consistency

- Same axes and visual grammar across panels.

Comparison

- Group patterns become visible without overplotting.

Scale

- Use free scales only when comparison is not the goal.

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Colour should encode, not decorate

1

Categorical — Distinct hues for unordered groups.

2

Sequential — Light-to-dark progression for ordered magnitude.

3

Diverging — Two directions around a meaningful midpoint.

4

Accessibility — Check contrast and add labels, shapes or facets.

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Annotation guides attention with evidence

1. **Direct**
   Label the important point or change.

2. **Reference**
   Add target, benchmark or event line.

3. **Restraint**
   Annotate the finding, not every mark.

4. **Source**
   Keep period, population and unit visible.

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Themes improve consistency without hiding data

1. **Hierarchy**
   Title, plot, annotation and source have distinct weight.

2. **Typography**
   Readable sizes and few font styles.

3. **Grid**
   Use light guides only where they help estimation.

4. **Whitespace**
   Separate elements and reduce cognitive load.

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Misleading visuals change the argument

CONCEPT

1. Truncated bar axis: Exaggerates small category differences.
2. Dual axes: Can manufacture apparent relationships.

CONTROL / DECISION

1. 3D: Distorts position, length and area.
2. Cherry-picking: Selective periods or groups hide contradictory evidence.

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Visual QA is an analytical control

Reconcile

- Displayed values agree with the source summary.

Inspect

- No clipping, overlap, wrong labels or hidden categories.

Challenge

- Test alternative scale, grouping or period choices.

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Interpret findings in five parts

1. Pattern — What is visibly happening?
2. Magnitude — How large is it and in what unit?
3. Uncertainty — What variability or evidence limits confidence?
4. Relevance — Why does it matter to the research decision?
5. Limit — What conclusion is not supported?

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Build Layered Quantitative Visuals

LAB 9

Use the grammar of graphics to create comparison, distribution and relationship plots with truthful scales, useful labels and accessible emphasis.

EXPECTED OUTPUT

A bar chart, faceted distribution plot and annotated scatter plot with a chart-choice rationale.

TOOLS R, ggplot2 when available, base-R fallback, scales, facets, annotation

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Build Layered Quantitative Visuals

ACCEPTANCE CHECK

All three plots answer their stated questions, remain legible without colour alone, use truthful scales and carry interpretations supported by the displayed data.

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Capstone: Review and Communicate Findings

LAB 10

Integrate preparation, summary, modelling, text evidence and visualisation into a compact reproducible healthcare research briefing.

EXPECTED OUTPUT

An auditable analysis bundle with control log, statistical results, three decision visuals and a concise evidence-led recommendation.

TOOLS R, base statistics, optional ggplot2, visual QA, reproducibility and interpretation

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Capstone: Review and Communicate Findings

ACCEPTANCE CHECK

The bundle reruns from source data, all controls pass or are explained, visuals agree with statistical outputs, and the recommendation distinguishes evidence from judgement.

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TOPIC RECAP

Recap — Data Visualization

1. **Knowledge**
   Question-to-chart fit and related principles.

2. **Ability**
   LO5

3. **Evidence**
   A validated workbook and decision-ready interpretation.

4. **Next**
   Connect this topic to the next layer of the analytical workflow.

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99

WRAP-UP

Evidence & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1

Prepare and transform data for analysis using R.

2

LO2

Summarise data with descriptive statistics, figures, graphs and tables.

3

LO3

Perform quantitative data analysis and interpret the findings.

4

LO4

Perform qualitative data analysis and determine significance or effect.

5

LO5

Create effective data visualisations in R and interpret research findings.

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BEFORE YOU LEAVE

Course Feedback

1. Complete the course feedback form
2. Rate the content and R labs
3. Rate the trainer and delivery
4. Share one improvement idea
5. Receive your certificate of completion

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THANK YOU

Turn research data into a defensible finding.

Prepare, summarise, analyse, visualise and interpret with evidence another analyst can reproduce.

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